MIL-C-43424A
22 April 1969

SUPERSEDING
MIL-C-43424
11 April 1966

MILITARY SPECIFICATION

CLOTHING, COMBAT TYPE, CONTOUR PACKAGING AND PACKING OF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the contour packaging and packing of combat clothing for storage and shipment.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-B-131	-	Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 101	-	Preservation, Packaging, and Packing Materials: Test Procedures.

FSC 8400

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 Contour packaging.- Contour packaging of combat clothing shall be level A.

3.1.1 Level A.-

3.1.1.1 Material.- The material used to effect the contour package shall be fabricated by combining scrim (cloth) backing, 0.001 inch thick polyethylene film, 0.0005 inch thick aluminum foil, and 0.0025 inch thick polyethylene film. The finished barrier material shall meet the test requirements for the class 1 barrier material of MIL-B-131.

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3.1.1.2 Bags.- The bags shall be formed from two sheets or by folding one sheet of the barrier material specified in 3.1.1.1, by placing heat sealable surfaces together and heat sealing along three edges leaving one edge open. The width of all heat seals, including the final closure of the bag, shall be not less than 3/8 inch, shall be continuous, and shall have a seal strength of not less than 7 pounds per inch of width when tested in accordance with 4.5.1. The edges of the side seams of the fabricated bag shall be parallel and all seams shall be parallel to the outer edges of the formed bag.

3.1.1.3 Contour packaging.- The folded size, when applicable, of each item and the quantity per contour package shall be as specified in table I. When the quantity per package is two or more of an item, the individual items shall be evenly stacked with every other one (or pair) reversed. The unit package quantity of an item shall be inclosed in the bag specified in 3.1.1.2. Closure of the bag shall be effected by heat sealing. Prior to final closure, the air within the bag shall be evacuated. The extent to which the air in the bag is evacuated shall be such that 24 hours after final closure the sealed package shall exhibit no internal pressure when tested in accordance with 4.5.2 or shall have a gage vacuum of 15 inches of mercury or greater on the basis of a barometric pressure of 30 inches of mercury when tested in accordance with 4.5.3. The finished package shall be rectangular in shape and shall exhibit no excessive bunching of material at the seams. Excess bag material, resulting from the removal of air from the package, shall be carefully folded back on the sealed package.

3.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

* 3.2.1 Level A.- Contour packages, specified in 3.1, in the quantity and arrangement specified in table I, shall be packed in a snug-fitting fiberboard box conforming to style RSC-L, V2s of PPP-B-636. The fiberboard box shall be equipped with liner and top and bottom pads made of V3c fiberboard conforming to PPP-F-320. The height of the liner shall equal the inside depth of the box less the thickness of the top and bottom pads. The size of the top and bottom pads shall be not more than 1/16 inch less than the inside length and width dimensions of the box. Each box shall be constructed, closed and reinforced with tape, banding or strapping in conformance with the box specification.

3.2.2 Level B.- Contour packages, specified in 3.1, in the quantity and arrangement specified in table I, shall be packed in a snug-fitting fiberboard box conforming to style RSC-L, type CF or SF, class domestic, variety SW, grade 275 of PPP-B-636. The fiberboard box shall be equipped with liner

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and top and bottom pads made of the same fiberboard as the box. The height of the liner shall equal the inside depth of the box less the thickness of the top and bottom pads. The size of the top and bottom pads shall be not more than 1/16 inch less than the inside length and width dimensions of the box. Each box shall be closed in accordance with method II of the appendix of the box specification.

* 3.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-F-320.

3.2.3 Level C.- Contour packages, specified in 3.1, in the quantity and arrangement specified in table I, shall be packed in shipping containers in a manner as to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 3.3 Palletization.- Unless otherwise specified (see 6.2), the shipping container shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

* 3.4 Marking.- In addition to any special marking required by the contract or order, unit packages, shipping containers, and unit loads shall be marked in accordance with the requirements of MIL-STD-129.

* 3.5 Workmanship.- The finished package shall conform to the quality and grade of product established by this specification.

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TABLE I.- Items for contour packaging

Items	Specification	Folded size 1/ (inches)	Unit pkg qty	Number of pkgs. per shipping container	Arrangement 2/ L	W	D
Blanket, Bed (Wool)	MIL-B-844	23 x 14-1/2	1	10 (on edge)	10	1	1
Coat, Man's, Combat, Tropical	MIL-C-43199	23 x 14-1/2	5	35	-	-	-
Coat, Man's, Field, M-65	MIL-C-43455	23 x 14-1/2	5	15	-	-	-
Drawers, Men's Cotton and Polyester/ Cotton, Boxer Style	MIL-D-40099	11 x 7-1/4	3	120	-	-	-
Drawers, Man's Winter Lightweight	MIL-D-43261	14-1/2 x 11-1/2	1	10 (one edge)	10	1	1
Glove Inserts Wool M-1949	MIL-G-835	---	2 pairs	36 (flat)	6	1	6
Hats and Insect Net; Hat, Comouflage Pattern (ERDL)	MIL-H-43577	14 x 4	1	144	-	-	-

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TABLE I.- Items for contour packaging (cont'd)

Items	Specification	Folded size 1/ (inches)	Unit pkg qty	Number of pkgs. per shipping container	Arrangement 2/ L	W	D
Liner, Coat Man's Field M-65	MIL-L-43536	23 x 14-1/2	1	10 (flat)	2	1	5
Liner, Parka Man's Nylon Quilted OG-1066	MIL-L-43466	23 x 14-1/2	1	10 (flat)	2	1	5
Liner, Poncho	MIL-L-43335	23 x 14-1/2	1	10	-	-	-
Liners, Trouser Man's Field, Nylon Quilted OG-106	MIL-L-43498	23 x 14-1/2	1	10 (flat)	2	1	5
Mitten Inserts Trigger Finger OG-208	MIL-M-809	---	2 pairs	36 (flat)	6	1	6
Scarf, Neckwear, Wool	MIL-S-2063	14-1/2 x 4-1/2	5	10 (flat)	2	1	5
Shirt, Man's Field Wool/ Nylon OG-108	MIL-S-10858	15 x 11-1/2	2	6 (flat)	2	1	3
Socks, Men's, Wool, Cushion Sole, Stretch Type	MIL-S-48	1/	5 pairs	20 (flat)	4	1	5

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TABLE I.- Items for contour packaging (cont'd)

Items	Specification	Folded Size 1/ (inches)	Unit pkg qty	Number of pkgs. per shipping container	Arrangement 2/ L	W	D
Sock, Men's, Wool, Winter	MIL-S-405	1/	5 pairs	20 (flat)	4	1	5
Trousers, Men's Combat, Tropical	MIL-T-43217B	23 x 14-1/2	5	35	-	-	-
Trousers, Men's Field, Wool, M - 1951	MIL-T-1870	23 x 14-1/2	1	10 (on edge)	10	1	1
Undershirt, Man's (Cotton, Quarter-Sleeve)	JJ-U-513	11 x 7-1/4	3	120	-	-	-
Undershirts, Men's Winter, Lightweight	MIL-U-43262	14-1/2 x 11-1/2	1	10 (on edge)	10	1	1

1/ Folding patterns shall follow descriptive method contained in specification for item except that folded sizes may be varied to obtain a more efficient final package size.

2/ L = Length W = Width D = Depth

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Examination of components.-

4.3.1 Examination of bags.- Examination of bags to determine compliance with 3.1.1.2 shall be in accordance with the classification of defects in table II. The sample unit for this examination shall be one bag. The lot size shall be expressed in units of bags. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 for major and 10.0 for total, expressed in terms of defects per hundred units.

TABLE II.- Examination of bags

Examine	Defect	Category Major	Minor
Bag material	Not combination specified.	101	
Bag construction	Seams not heat sealed through entire length of seam.	102	
	Seams less than 3/8 inch wide.	103	
	Seams not parallel to outer edges of bags.		201
	Side seams not parallel to each other.	104	

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4.4 Inspection of the end item.-

4.4.1 Examination of the end item.- Examination shall be made in accordance with the classification of defects in tables III and IV. The sample unit for this examination shall be one pack. The lot size shall be expressed in units of packs. The inspection level shall be S-2 and the AQL shall be 2.5 for major and 10.0 for total, expressed in terms of defects per hundred units.

TABLE III.- Examination of packaging

Examine	Defect	Category Major	Minor
Package	Item not folded as specified (see table I).	101	
	Quantity per package differs from that specified in table I.	102	
	Unevenly stacked.	103	
	Alternate items or pairs, as applicable, not reversed in stack.		201
	Closure of bag not heat sealed.	104	
	Excess air in bag.	105	
	Package not rectangular in shape.		202
	Excessive bunching of material at seams.	106	
	Excess bag material not folded over package.		203

TABLE IV.- Examination of packing and marking

Examine	Defect	Category Major	Minor
Pack	Packages do not make snug fit in container.		201
	Liner missing.	101	
	One or more pads missing.	102	
	Liner and pad material differs from material specified.		202
	Size of liner and pads not as specified.	103	
Marking	Omitted; incomplete; incorrect; illegible; of improper size, location, sequence, or method of application.	104	

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4.4.2 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for closure and strapping shall be in accordance with the appendix of that specification.

* 4.4.3 Examination for palletization.- An examination shall be made to determine that the palletization requirements specified in 3.3 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Palletized pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

* 4.4.4 Testing of the end item.- The end item shall be tested for applicable characteristics as indicated in table V. The sample unit shall be three bags.

TABLE V.- Instructions for testing

Characteristic	Basic req. para.	Test para.	Results reported to	No. of determinations per unit	Inspection level 1/	AQL
Heat seal seam test	3.1.1.2	4.5.1	Nearest lb. per inch width	2/	S-2	2.5
Vacuum retention test	3.1.1.3	4.5.2	Nearest inch	1	S-2	1.5
Vacuum test	3.1.1.3	4.5.3	Nearest inch	1	S-2	1.5

1/ Lot size shall be one day's production.

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2/ Report average of test from each seam of one bag.

4.5 Tests.-

4.5.1 Heat sealed seam test.- Seam tests shall be conducted in accordance with method of 246 of Fed. Std. 101. Sections of the heat sealed seams shall be obtained from sealed packages. Sections of the heat seals 0.5 inch in width shall be cut perpendicular to the line of the seal. The length of the legs of the test section is not critical. The sections shall be unfolded, and with the seal positioned midway between the jaws, the legs of the section shall be clamped in the tensile testing machine. The load shall be applied perpendicular to the line of the seal.

* 4.5.2 Vacuum retention test.- Each sample package shall be subjected to the vacuum chamber technique described in method 241 of Fed. Std. 101 except that no water will be required in the chamber. The vacuum in the chamber shall not be less than 15 inches of mercury.

* 4.5.3 Vacuum test.- The package shall be allowed to reach a temperature of 75° ± 5°F. The vacuum reading shall then be taken with a puncture type vacuum gage. The vacuum gage used shall be scaled to provide a reading in inches of mercury.

5. PREPARATION FOR DELIVERY
(Not applicable).

6. NOTES

6.1 Intended use.- Contour packaging is intended for application to soft, compressible items for the purpose of reducing the bulk and for protection of clothing and textile items against deterioration during indeterminate periods and conditions of storage.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of applicable item specification.
c. Selection of the applicable levels of packing (see 3.2).
d. When weather-resistant shipping containers are required for level B shipment (see 3.2.2.1).
e. When palletization is not required (see 3.3).

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* 6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 82

Preparing activity:

Army - GL

Project No. 8405-0228

Review activities:

Army - SM
Navy - SA
Air Force - 82, 85

User activity:

Navy - MC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Clothing, Combat Type, Contour Packaging and Packing Of MIL-C-43424A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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